23009145 6 王柯贺

1、数值误差的避免

(a) $\Delta(x_1+x_2) = (x_1+x_2)\frac{\varepsilon}{2}$

$\Rightarrow \Delta S = (x_1+x_2)\frac{\varepsilon}{2} + (x_1+x_2+x_3)\frac{\varepsilon}{2} + \cdots + (x_1+x_2+\cdots+x_N)\frac{\varepsilon}{2} \le \frac{N^2}{2}\bar{x}\varepsilon$

$\Rightarrow \Delta\bar{x} = N\bar{x}\frac{\varepsilon}{2}$

(b) ① 设 $\sum_{i=1}^{K} x_i^2 = S_{2K}$

$\Rightarrow S^2 = \frac{1}{N-1}\left\{(1+\frac{\varepsilon}{2})\left[S_{2N} + S_{22}\frac{\varepsilon}{2} + \cdots + S_{2N}\frac{\varepsilon}{2}\right] - N\bar{x}^2(1+\frac{\varepsilon}{2})(1+N\frac{\varepsilon}{2})^2\right\}$

$\Rightarrow \Delta S^2 = \frac{1}{N-1}\left\{\frac{\varepsilon}{2}S_{2N} + (S_{22}+\cdots+S_{2N})\frac{\varepsilon}{2} + N\bar{x}^2(\frac{\varepsilon}{2}+N\varepsilon)\right\}$

$\sim \frac{1}{N-1}\left\{\frac{\varepsilon}{2}\cdot N^2\bar{x}^2\right\}$

② $S^2 = \frac{1}{N-1}\sum_{i=1}^{N}\left[(x_i - \bar{x}(1+N\frac{\varepsilon}{2}))(1+\frac{\varepsilon}{2})\right]^2$

$\Delta S^2 \approx \frac{1}{N-1}\sum_{i=1}^{N}\left[(x_i-\bar{x})\cdot N\bar{x}\varepsilon + (x_i-\bar{x})^2\varepsilon\right]$

$\approx \frac{1}{N-1}\left[|x_i-\bar{x}|\bar{x}N^2\varepsilon\right]$

第二种好、有 $|x_i-\bar{x}|$ 压低误差项

(3) $I_n = \int_0^1 dx\left(\frac{x^n}{x+5}\right)$

$I_0 = \int_0^1 dx\ \frac{1}{x+5} = \ln\frac{6}{5}$

$I_n + I_{n-1}\times 5 = \int_0^1 dx\ \frac{x^{n-1}(x+5)}{x+5} = \int_0^1 x^{n-1}dx = \frac{1}{n}$

$I_n = (\frac{1}{n} - 5I_{n-1})$

$= \frac{1}{n} - 5(\frac{1}{n-1} - 5I_{n-2})$

$= \sum_{i=2}^{n}\frac{(-1)^{n+i}5^{(n-i)}}{i} + (-5)^{n-1}I_1$

$\Rightarrow$ ε会被放大 $(-5)^{n-1}$ 倍，不稳定

2.

(1) $A = \begin{pmatrix} 1 & -1 & \cdots & -1 \\ & 1 & \ddots & \vdots \\ & & \ddots & -1 \\ & & & 1 \end{pmatrix}$ 第n列 + = 第一列，可消去第一行 -1

类似有 $\det(A) = \det(I) = 1$

(2) $Ax^1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ $x^1 = \begin{pmatrix} x_1^1 \\ \vdots \\ x_n^1 \end{pmatrix}$

$\Rightarrow \begin{cases} x_n = 0 \\ x_{n-1} = x_n \\ x_{n-2} = x_n + x_{n-1} \\ \vdots \\ x_1 = x_2 + \cdots + x_n + 1 \end{cases}$ $\Rightarrow x^1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ 类似 $x^2 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ $x^n = \begin{pmatrix} x^{2n} \\ \vdots \\ 4 \\ 2 \\ 1 \end{pmatrix}$

$\Rightarrow A^{-1} = \begin{pmatrix} 1 & 2 & 4 & & 2^n \\ & 1 & 2 & & \vdots \\ & & 1 & & \vdots \\ & & & \ddots & 1 \\ & & & & 1 \end{pmatrix}$

(3) $\|A\|_p = \sup \frac{\|Ax\|_p}{\|x\|_p}$

$\lim_{n\to\infty} \sqrt[n]{\sum_{i=1}^{N} x_i^n} = \max\{i=1,\cdots N \mid |x_i|\}$

$\Rightarrow \|A\|_\infty = \sup \frac{\max(Ax)}{\max(x)}$

$= \sup \frac{\max_j(|\sum_i A_{ji} x_i|)}{\max_i(|x_i|)}$ 若取 $x_i = \pm\frac{1}{\sqrt{N}}$ 满足 $x_i A_{ij} = |x_i A_{ij}|$

则有最大 $\|A\|_\infty = \max_j\left(\sum_i |A_{ji}|\right)$

(4) 么正矩阵 $U^+ = U^{-1}$

$\|U\|_2 = \sup \frac{\|Ux\|}{\|x\|}$ 由于 $\langle Ux, Ux\rangle = \langle U^+Ux, x\rangle = |x|^2$

$\Rightarrow \|U\|_2 = 1 = \|U^+\|_2$

类似 $\|UA\|_2 = \sup \frac{\|UAx\|}{\|x\|}$ 又 $\|UAx\| = \|Ax\|$

$= \|A_2\|$

3.

(a) $B = A^T A = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 9 & 8 \\ 8 & 9 \end{pmatrix}$

$(9-\lambda)^2 = 64$

$9-\lambda = \pm 8 \quad \lambda = 9 \pm 8 = 17$ or 1

$\begin{pmatrix} 9 & 8 \\ 8 & 9 \end{pmatrix} - 17 \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} = \begin{pmatrix} -8 & 8 \\ 8 & -8 \end{pmatrix}$

$\Rightarrow v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\begin{pmatrix} 9 & 8 \\ 8 & 9 \end{pmatrix} - 1 \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} = \begin{pmatrix} 8 & 8 \\ 8 & 8 \end{pmatrix}$

$\Rightarrow v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \Rightarrow V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

证明：$B = A^T A \Rightarrow B^T = A^T A = B$

$\Rightarrow$ 设 λ 为 B 本征值 则 $Bv = \lambda v$

有 $\langle v, Bv \rangle = \lambda \|v\|^2 = \langle Bv, v \rangle = \lambda^* \|v\|^2$

$\Rightarrow \lambda = \lambda^*$ 本征值为实数.

又有 对于所有本征向量 v $\langle v, Bv \rangle = \langle Av, Av \rangle = \|Av\|^2 = \lambda \|v\|^2$

$\Rightarrow \lambda \geq 0$ 又 对不同、本征向量 v, w，设 $Bv = \lambda_1 v$ $Bw = \lambda_2 w$

$\Rightarrow \langle v, Bw \rangle = \lambda_2 \langle v, w \rangle = \lambda_1 \langle v, w \rangle$ 当 $\lambda_1 \neq \lambda_2$ 时 $\langle v, w \rangle = 0$ 正交

归纳法证明 特征向量可张满整个空间.

故 $A^T A$ 可对角化，且特征值非负

(b)

$u_1 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ 2 & 2 \end{pmatrix} \frac{1}{\sqrt{34}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{34}} \begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix} \quad u_3 = u_1 \times u_2 = \frac{1}{\sqrt{17}} \begin{pmatrix} 2 \\ 2 \\ -3 \end{pmatrix}$

$u_2 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ 2 & 2 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

(c)

$$A_1 = \sqrt{17}\ \frac{1}{\sqrt{34}}\begin{pmatrix}3\\3\\4\end{pmatrix}\frac{1}{\sqrt{2}}(1,1) = \frac{1}{2}\begin{pmatrix}3&3\\3&3\\4&4\end{pmatrix}$$

$$A - A_1 = \begin{pmatrix}1/2 & -1/2\\ -1/2 & 1/2\\ 0 & 0\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1 & -1\\ -1 & 1\\ 0 & 0\end{pmatrix}$$

$$\|\Delta A\|_F = 1 = \sqrt{\lambda_2}$$

4. $$x^T A x = \sum_{i,j} a_{ij} x_i x_j$$

$$= \sum_i a_{ii} x_i^2 + \sum_{i\neq j} a_{ij} x_i x_j$$

$$> \sum_i \sum_{j(j\neq i)} |a_{ij}|\, x_i^2 + \sum_{\substack{i,j\\(i\neq j)}} a_{ij} x_i x_j$$

$$= \sum_{\substack{i,j\\(i\neq j)}} |a_{ij}|\left(x_i^2 + \operatorname{sgn}(a_{ij})\, x_i x_j\right)$$

$$= \sum_{\substack{i>j\\(i\neq j)}} |a_{ij}|\left(x_i^2 + x_j^2 + 2\operatorname{sgn}(a_{ij})\, x_i x_j\right)$$

$$= \sum_{i>j} |a_{ij}|\left(x_i + \operatorname{sgn}(a_{ij}) x_j\right)^2 \geq \sum_{i=j} |a_{ii}|\, 4x_i^2 > 0$$

5.

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h= [1.e-01 1.e-02 1.e-03 1.e-04 1.e-05 1.e-06 1.e-07 1.e-08 1.e-09 1.e-10
 1.e-11 1.e-12 1.e-13 1.e-14 1.e-15 1.e-16 1.e-17 1.e-18 1.e-19 1.e-20]
f'= [0.0437105  0.03454619 0.0337741  0.03369816 0.03369058 0.03368982
 0.03368974 0.03368974 0.03368973 0.03368974 0.03368972 0.0336866
 0.033697   0.03348016 0.03642919 0.06071532 0.         0.
 0.         0.        ]
```

此为 h 与 f' 关系，当 $h = 10^{-17}$ 时精度明显下降

numpy 计算 np.exp 的算法大概是 $\exp(x) = 2^{\mathrm{round}(x/\ln 2)} \times e^{x - \frac{x}{\ln 2}}$

当 $x = -1$ 时，$x - \frac{x}{\ln 2} \approx 0.3068$. python 双精度浮点数精度只有 $10^{-16} \sim 10^{-17}$

在这之后，计算不准确.

6.

直接使用khan算法循环了1e7次，然后和mathematica 计算的结果进行比较，